

Locomotion-Grounded Soccer Skills: A Unified Humanoid Policy for Free Navigation and Multi Motion-Guided Kicking Skills

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Abstract—Recent humanoid soccer systems obtain whole-body striking ability by making motion tracking the substrate of the controller and deriving locomotion from it, typically by steering a motion-reference anchor toward the ball. This yields strong shooting results, but locomotion is then only ever trained on the narrow, deterministic command distribution that ball approach induces, and is never evaluated as a capability in its own right. We invert the stack. A general, command-conditioned locomotion policy is trained first as a substrate, and N motion-guided kicking skills are added on top of it as task-gated layers, so that the reachable gait space is set by the locomotion command curriculum rather than by the content of any reference clip. Because every skill both initiates from and terminates into the same commandable locomotion state, locomotion also acts as a composition hub: a library of N skills requires $O(N)$ transition specifications rather than $O(N^2)$ pairwise ones, and post-strike stabilization is a handoff back to a trained controller rather than a scripted per-clip recovery reference. We instantiate the architecture on a 29-DoF Unitree G1 with a library of seven retargeted kicking skills whose nominal aim directions span 259.5° , including lateral, rearward and weak-foot strikes that a single forward-facing reference does not express. On [N] evaluation conditions we report **[headline shooting result]**, alongside command-tracking, terrain and push-recovery measurements taken *with the full skill library attached*, an axis that, to our knowledge, prior humanoid soccer systems do not report.

I. INTRODUCTION

Soccer has become a demanding benchmark for humanoid whole-body control because it stresses two capabilities at once, on the same hardware and under a single controller: sustained, general locomotion (repositioning, retreating, sidestepping, turning in place), and precise, contact-rich strikes whose entire useful signal arrives in a sub-10 ms impulse with delayed feedback about where the ball went. Reinforcement learning has recently advanced both halves [1]–[7], and the strongest reported shooting result on a full-size humanoid comes from motion-guided curriculum learning: [2] attains 0.73 m mean target error from 3 m at peak ball speeds of 13.10 m/s on a Unitree G1.

The recipe behind these results is consistent across systems: a human kicking clip is retargeted to the robot [8], [9], a tracking reward supplies the whole-body coordination pure task rewards struggle to discover [10]–[12], and ball and target rewards are layered on top. Locomotion enters last and indirectly, by steering the tracking anchor toward a predicted ball position while the tracking reward stays active throughout approach [2]: motion tracking is the *substrate*, and walking is a behavior grown on top of it. We adopt much

of this reward machinery, but the ordering has a consequence the literature has not measured. Locomotion is only ever exercised during ball approach, so the training command distribution is whatever that approach heuristic emits. [2]’s deterministic anchor-steering law issues no lateral, backward, or in-place-rotation command, and no command-tracking, rough-terrain, push-recovery or omnidirectional result is reported there or in [3], [4]. This is an observation about training and evaluation, not architecture: Stage-3 locomotion in [2] is already reward-driven, and widening its command distribution is a curriculum change, not an architectural one. The gap is empirical: general locomotion in a humanoid soccer policy is neither trained across a broad command envelope nor reported as a measured capability, and closing it is what this paper does.

A second consequence follows from the same design: training against a single reference kick bounds the strike repertoire to one contact type. [2] report a natural right-footed bias, degradation at extreme lateral and high targets, and name multi-skill priors as their principal open problem; their simulated targets sit on a goal plane 5 m ahead, entirely within the forward hemisphere. A match calls for a *library*: an inside-foot pass, an instep drive, a lateral flick, a strike behind the current heading.

Approach. We invert the stack (Fig. 1). A general, command-conditioned locomotion policy is trained first, with a randomized (v_x, v_y, ω_z) envelope, mixed terrain, and push, mass, friction and actuation randomization, and with no ball or reference clip present at all. Only then are N motion-guided kicking skills added as task-gated layers: a task-mode variable hard-zeroes every kick objective in locomotion mode rather than discounting it to a residual weight (Fig. 2). The reachable gait space is therefore set by the locomotion curriculum alone, independent of what any clip contains.

Placing locomotion at the base also turns it into a composition hub: since every skill starts from and returns to the same commandable state, a library of N skills needs $O(N)$ transitions rather than $O(N^2)$ pairwise ones. It also changes what happens after contact. Prior work tracks a synthetic, per-clip stabilization reference for a fixed window after the strike, a load-bearing but momentum-blind component ([2] reports alive rate falling from 98.8% to 24.4% without it); we instead hand control straight back to the trained locomotion controller, already robust to disturbance by construction.

We instantiate this on a 29-DoF Unitree G1 at 50 Hz, training in IsaacLab [13] with an off-policy maximum-entropy learner [14], [15] and merging per-skill specialists into one deployable network by DAgger-style distillation [16], [17]

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(Fig. 2). The resulting library spans 259.5° of nominal aim direction at 4.0–7.2 m strike distances, including directions no single forward-facing reference expresses. We evaluate in MuJoCo [18] sim-to-sim on the exported deployment policy, and on hardware.

Contributions.

- **A locomotion-grounded architecture for humanoid soccer.** A command-conditioned locomotion policy is the substrate and motion-guided kicking skills are task-gated layers above it, with kick objectives hard-zeroed outside kick mode. We show the skill library can be attached without measurable degradation of the locomotion base ([Table ref]).
- **A multi-skill strike library with quantified directional coverage.** Seven retargeted skills spanning 259.5° of nominal aim direction, including lateral, rearward and weak-foot strikes. We report per-skill and union coverage, gaps included, rather than a single forward-cone accuracy number.
- **A learned kick→locomotion handoff** in place of a scripted per-clip stabilization tail, together with an interruptibility analysis measuring survival when the skill is aborted at an arbitrary tick of the swing.
- **A joint evaluation of locomotion generality and shooting quality in one policy.** We report command tracking across the full envelope, terrain traversal and push recovery measured with the skill library attached, alongside standard shooting metrics. Prior humanoid soccer systems do not report this combination.

We are explicit about what the architecture does not yet deliver. Skill selection is commanded, not learned. Directional coverage is *not* continuous: the seven skills tile only 155.8° across five islands, leaving four gaps (largest 59.4°) that we report rather than smooth over. And entering a skill is a weaker claim than leaving one: return-to-locomotion is robust at any tick, but entry from motion survives while losing accuracy in proportion to entry speed, and every entry we measure is handed a correctly placed ball, so walking up to a stationary one under command is a step we have not established. Section [VII] treats each directly.

II. RELATED WORK

A. Humanoid Soccer and Robot Sports

Reinforcement learning has produced athletic, contact-rich sports behaviors across embodiments, including quadrupedal soccer [19], humanoid table tennis [20] and badminton with legged manipulators [21]. On bipeds, [1] demonstrated end-to-end one-versus-one soccer on a miniature platform, with a behavior repertoire spanning walking, turning, kicking and getting up, and transitions learned rather than scripted. Their pipeline trains separate teacher policies and distills them into a single feed-forward network, which is also the structure we adopt for merging skills; the deployed policy is conditioned on egocentric game state (ball, opponent and goal), rather than exposing a velocity-command interface, and learns a single scoring strike.

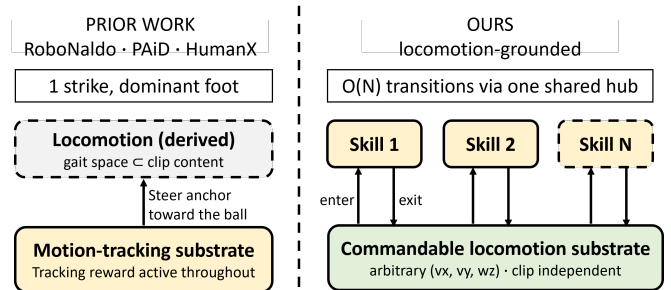


Fig. 1: **Substrate inversion.** Prior motion-guided soccer systems make motion tracking the substrate and obtain locomotion from it by steering the tracking anchor toward the ball, with the tracking reward active throughout approach. We place a commandable locomotion policy at the base instead and attach N task-gated skills above it, so every skill enters from and exits into one shared hub state and the library needs $O(N)$ transitions.

On full-size humanoids, a rapidly growing line of work targets shooting quality directly. [22] optimizes foot trajectories for controlled kicking without learning; [7] addresses the goalkeeping side under position-conditioned task-motion constraints; [6] and [4] study vision-driven and noisy-perception striking respectively; [3] proposes a progressive perception-action framework; and [5] learns generalizable interaction skills from human video. [2] reports the strongest shooting numbers to date via a three-stage motion-guided curriculum (clip imitation, then ball and target rewards, then a locomotion command interface), and is the closest point of comparison for this work. We adopt substantial parts of its reward design for the strike itself. We differ in the ordering of the curriculum, in training locomotion against a randomized command envelope rather than a ball-approach heuristic, in carrying a library of seven strike types rather than one reference, and in returning post-strike control to a trained controller rather than a scripted stabilization reference. Across this literature, we are not aware of a system that reports locomotion metrics independent of ball approach, nor one that reports more than a single strike contact type. Table I summarises these systems along the axes this paper targets. Two columns, omnidirectional locomotion and multiple strike types, are unfilled for every prior system, and they are the two the architecture in this paper is designed to occupy.

B. Motion Tracking and Imitation for Whole-Body Control

Reference-guided RL supplies whole-body coordination that task rewards alone struggle to discover, from example-guided character control [10] to adversarial motion priors that match a distribution of behaviors rather than a single trajectory [11]. Recent humanoid systems track retargeted human motion directly on hardware [12], supported by video-based motion recovery [8] and general-purpose retargeting [9]; we use the same acquisition pipeline for every clip in our library. The known tension is that tracking constrains timing and contact selection at the same time as it supplies stability,

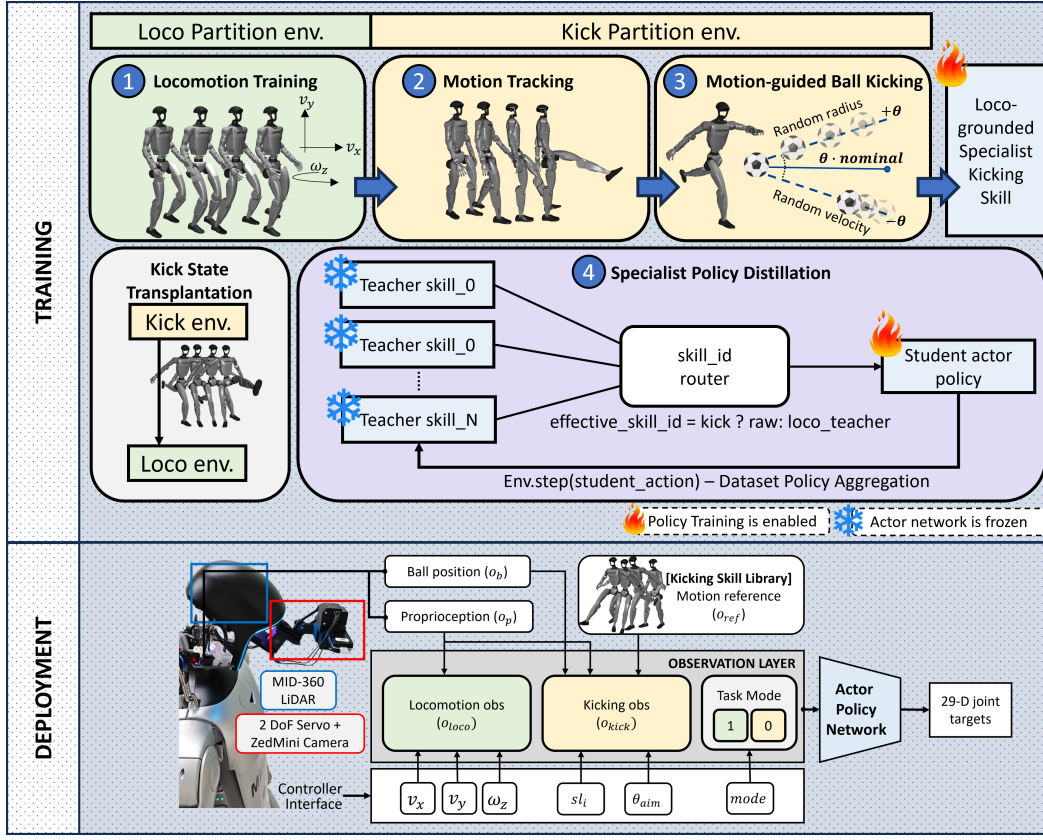


Fig. 2: **Training pipeline and deployed architecture.** **Training (top).** (1) A commandable locomotion policy is trained first on the locomotion partition, with no ball and no reference clip. (2) Kick-partition environments add motion tracking of a retargeted human clip. (3) Ball and target rewards are added, with the strike direction commanded as an aim angle θ about the skill’s nominal bearing and randomisation over ball placement and speed, yielding one locomotion-grounded specialist per clip. (4) The specialists are merged into a single deployable actor by DAgger-style distillation: frozen teachers label states that the *student* visits, routed by skill index, with locomotion environments defaulting to the locomotion teacher. Kick state transplantation injects live kick-environment states into locomotion environments at reset, training recovery from kick-disturbed poses. **Deployment (bottom).** The same distilled network deployed: a task-mode variable gates the observation into locomotion and kicking groups before a single actor network that emits 29-D joint position targets at 50 Hz. The controller interface exposes velocity commands (v_x, v_y, ω_z) for locomotion, and a skill index sl_i with an aim angle θ_{aim} for kicking. [Perception hardware shown at left is used for hardware demonstrations only; scope this in Sec. [V] or drop it from the figure.]

which is precisely what staged curricula are designed to relax [2]. Our task-mode gating takes that relaxation to its limit in one direction: the tracking objective is not merely discounted outside the skill, it is switched off, so that locomotion is optimized against its own reward specification with no residual clip term shaping it.

C. Commandable Locomotion as a Primitive

Massively parallel simulation has made velocity-commanded legged locomotion with terrain and disturbance curricula a well-established primitive [13], [23], and recent work has pushed the wall-clock cost of obtaining a sim-to-real-capable humanoid locomotion policy to minutes using off-policy maximum-entropy learning [14], [15]. The resulting policies track arbitrary commands over rough terrain and recover from disturbances. Our architecture places this capability at the *base* of the stack, where it

is trained and evaluated on its own terms, rather than obtaining walking as a by-product of ball approach. We note that the contribution is learner-agnostic: the task gating, the locomotion hub and the handoff are properties of the problem formulation, not of the optimizer, and nothing in them requires off-policy learning.

D. Whole-Body Object Interaction

Contact-rich loco-manipulation systems typically assume dwell-time contact and accumulate interaction reward over many steps [24]–[27]. Shooting sits at the opposite end of that spectrum: a single brief impulse determines the outcome, and the reward signal that grades it arrives only after the ball has traveled. This is what motivates the specialized interaction and aiming rewards used here and in [2], and it is also why post-contact stability must be handled explicitly: the robot is left in a single-support, momentum-carrying

TABLE I: Capability comparison against prior humanoid soccer systems.

Method	Platform	Locomotion substrate		Skill repertoire			Handoff	
		Free cmd.	Omni dir.	Multi skill	Motion guided	Lateral tgt.	L→K	K→L
Haarnoja et al. [1]	Miniature	× [†]	○	× [‡]	×	○	✓	✓
STOFT [22]	Bipedal	×	×	×	×	×	×	×
Reactive [6]	Humanoid	○	×	○	✓	×	×	×
Striker [4]	Humanoid	○	×	×	×	○	×	×
Goalkeeper [7]	Humanoid	×	×	×	✓	×	×	×
HumanX [5]	Humanoid	×	×	✓	✓	×	×	×
PAiD [3]	Humanoid	○	×	×	✓	○	×	×
RoboNaldo [2]	Humanoid	○	×	×	✓	○	×	○
Ours	Humanoid	✓	✓	✓	✓	✓	○ [§]	✓

✓ = supported and demonstrated; ○ = mechanism present but restricted to the ball-approach regime, or not evaluated as a general capability; × = absent. *Free cmd.*: accepts arbitrary externally issued velocity commands, not only commands emitted by a ball-approach planner. *Omni dir.*: sustained lateral, backward and in-place rotation demonstrated. *Multi skill*: more than one distinct strike or contact type. *Motion guided*: skill reward shaped by tracking or matching human reference motion, rather than emerging from a task reward alone or from a non-learned trajectory. *Lateral tgt.*: demonstrated strike accuracy to off-axis, non-forward-cone targets. *Handoff*: entry into (L→K) and exit from (K→L) a skill realised as a transition to or from a trained locomotion controller, rather than a kick-tracking reward that stays active throughout approach, or a scripted stabilisation reference after contact. Marks reflect capabilities *as reported in each paper*; an unreported capability is scored × or ○ rather than assumed absent. [†]Conditioned on egocentric game state (ball, opponent, goal), with no velocity-command interface for a human or high-level planner. [‡]Exhibits a behaviour repertoire (walk, turn, kick, get-up) but a single strike type; this column tracks contact-type diversity specifically. [§]Our skill entry is a handoff from the trained locomotion controller, but currently requires first settling to a near-zero velocity state; see Sec. [VII].

configuration exactly when the task reward has already been collected.

E. Skill Libraries, Composition and Transitions

The options framework formalizes temporally extended actions through initiation sets, policies and termination conditions [28]. A library is composable when each skill's termination set is contained in the next skill's initiation set; a hub state that all skills share makes this condition hold by construction and reduces the number of transitions that must be specified or learned from $O(N^2)$ to $O(N)$. Our locomotion mode plays exactly this role. For obtaining a single network that executes the whole library, we use behavior distillation: independently trained specialists supply supervision to one student under the student's own state distribution [16], following the pattern used for policy merging in [17]. [1] pursues the same end in the soccer setting, merging independently trained skill policies into one feed-forward network, but by a different mechanism: their full policy is trained by reinforcement learning under self-play and kept near the skills by a KL regulariser, where ours regresses directly onto a routed teacher's actions with no reward term in the merging stage. We emphasize that a single deployed network is not by itself a distinguishing property of our system, since both [1] and [2] also deploy one non-hierarchical network. What differs is which capability that network is grounded in, and how many distinct strike types it carries.

III. PROBLEM FORMULATION

We model the problem as a task-conditioned Markov decision process solved by one policy. Each environment carries a task-mode variable

$$m_t \in \{\text{LOCO}, \text{KICK}_1, \dots, \text{KICK}_N\}, \quad (1)$$

which selects what the policy is asked to do and what it is rewarded and terminated for.

A. Task-mode gating

Rather than arbitrate between separate controllers, m_t gates three things inside one network. With $\mathcal{T}$ indexing reward terms and w_τ the weight of term τ ,

$$r_t = \sum_{\tau \in \mathcal{T}} g_\tau(m_t) w_\tau \hat{r}_\tau(s_t, a_t), \quad g_\tau(m_t) \in \{0, 1\}, \quad (2)$$

where g_τ is 1 only in the mode its term was authored for. The gate is a hard zero, not a reduced weight: in locomotion mode every kick term contributes nothing to the gradient, and conversely. This is what carries the substrate claim. Tracking rewards left active at reduced weight would still shape the reachable gait space; zeroing them means locomotion is optimised against its own specification alone. Terminations are gated identically.

The observation is gated by masking, not omission. Inactive-mode terms are zeroed but stay in the vector, so the input is 261-dimensional in every mode and one network serves both without a change of input shape (Table II). A task-mode one-hot stays live in both, so the policy is told its mode rather than inferring it from which blocks are zero.

B. Skill conditioning

Within kick mode a skill is specified two ways. Its retargeted reference trajectory enters the observation directly, making the policy reference-conditioned rather than index-conditioned: there is no learned skill embedding. The commanded direction enters separately as a scalar aim angle θ_{aim} , relative to that skill's nominal bearing and normalised by a fixed reference angle, so it is bounded in $[-1, 1]$ by construction and needs no robot localisation at deployment.

TABLE II: Actor observation space, 261 dimensions.

<i>Locomotion group (o_{loco})</i>		<i>Kicking group (o_{kick})</i>	
Term	Dim	Term	Dim
Base angular velocity	3	Motion reference ($q, \dot{q}$)	58
Projected gravity	3	Reference orientation	6
Command (v_x, v_y, ω_z)	3	Base angular velocity	3
Joint position	29	Joint position	29
Joint velocity	29	Joint velocity	29
Previous action	29	Previous action	29
Gait phase, sin	2	Ball position o_b	3
Gait phase, cos	2	Aim command θ_{aim}	2
Task-mode one-hot, active in both modes: 2			

Joint terms are 29-dimensional, one per actuated DoF; the motion reference concatenates reference joint position and velocity. The gait phase is two-dimensional because the two legs are held half a cycle apart. Reference orientation is the first two columns of the rotation matrix of the clip’s reference body expressed in the robot’s own reference-body frame. Projected gravity is the unit gravity vector expressed in the base frame, encoding roll and pitch relative to vertical while remaining invariant to heading; on hardware it is obtained directly from the onboard IMU orientation. Every actor term is measurable on the robot, which is why base linear velocity, not reliably observable without external odometry, appears only in the critic.

The width is constant across modes. Task-mode gating *zeroes* the inactive group rather than removing it, so the vector is 261-dimensional in every mode and one network serves both without a change of input shape. The one-hot tells the policy which mode it is in, so it never has to infer this from which blocks happen to be zero.

C. Handoff

A skill ends when its authored reference content is exhausted, m_t returns to LOCO, and the trained locomotion policy resumes control. We call this a handoff rather than a stabilisation phase because no synthetic recovery reference is tracked: the robot is handed back to a disturbance-robust controller in whatever single-support, momentum-carrying state the strike left it. The reverse transition is commanded rather than learned, and is where a precondition applies (Sec. VII).

Table II details the observation space of Fig. 2’s deployment view.

IV. METHOD

Training proceeds in the four stages of Fig. 2. The ordering is the claim: locomotion is trained first and alone, and everything else is layered onto it.

A. Stage 1: locomotion substrate

A command-conditioned locomotion policy is trained with no ball and no reference clip present, on commands sampled from a randomised (v_x, v_y, ω_z) envelope covering forward, backward, lateral and in-place rotation, over mixed terrain with disturbance randomisation. This stage fixes the reachable gait space, and with no clip present that space is a property of the command curriculum alone.

B. Stage 2: motion tracking

Kick-partition environments add a tracking objective against a human kick clip, recovered from video [8] and retargeted [9]. Ball and target rewards stay at zero weight here, so the policy learns the coordination of the strike

before it is asked to aim. Locomotion environments continue the unmodified Stage-1 task: this is an addition, not a replacement.

C. Stage 3: motion-guided ball kicking

The shooting objective is ramped linearly from zero over 5000 iterations, avoiding a step change at the stage boundary. The aim angle θ is sampled per attempt within $\pm 15^\circ$ of the skill’s nominal bearing, with randomised ball placement and speed. Table III gives the full reward specification; the largest weights sit on ball-to-target error and post-contact ball speed, and the alive reward is attenuated $10\times$ before contact so that standing still and declining to strike is not a viable optimum.

A separate mechanism supplies the boundary training the handoff depends on. Some locomotion environments are seeded at reset from a kick-disturbed pose rather than a standing one, taken either from the reference clip or, with higher probability, transplanted from the live state of a running kick environment. In the runs reported here this is enabled during distillation (Stage 4); the ablation retrains it at specialist level to isolate it. The second source is the faithful one: measured tracking error against the reference is never zero, and contact forces have no representation in motion capture at all. Training recovery from states the policy actually produces closes that gap at reset, without paying for a mid-episode flip. Table XI(b) isolates it.

Each skill is trained this way from the shared Stage-1 substrate, giving one locomotion-grounded specialist per clip.

D. Stage 4: specialist distillation

Specialists are merged into one deployable actor by DAgger-style distillation [16], [17]. The *student* acts in the environment and frozen teachers label the states it visits, which is what separates this from behaviour cloning: supervision covers the states the student drifts into, not the teachers’ trajectories. A skill-index router picks the teacher, with locomotion environments defaulting to a locomotion teacher.

Two details are load-bearing. Each teacher must run under its own frozen observation normaliser, since the wrong normaliser yields a confident wrong action and raises nothing. And the rollout action form must match the regression target; sampling one and regressing the other is self-amplifying in a DAgger loop, because the executed action determines which states are visited next.

E. Learning algorithm

All stages use FastSAC, an off-policy maximum-entropy actor-critic [14], [15], adopted because simulation throughput rather than environment count is the bottleneck in contact-rich training, so replay extracts more updates per simulated step. Running two tasks in one network motivates two changes: entropy target and discount factor are decoupled per task mode, so the horizon suited to a multi-second kick is not forced onto locomotion. The architecture is learner-agnostic; nothing in the gating, hub or handoff requires off-policy learning.

TABLE III: Kick-mode reward terms and weights.

Term	w	Term	w
<i>Motion tracking</i>			
Global root pos.	1.0	Global root orient.	0.5
Relative body pos.	1.0	Relative body orient.	1.0
Global body lin. vel.	1.0	Global body ang. vel.	1.0
Global feet lin. vel.	1.0	Strike joint null-sp.	1.0
Strike leg null-sp.	1.0		
<i>Shooting</i>			
Ball-to-target error	20.0	Ball velocity	10.0
Predicted ball-to-tgt.	5.0	Contact orientation	4.0
Ball proximity	2.0	CoM to ball dist.	2.0
Torso to ball dist.	2.0	Foot strike pitch	1.0
Ball over line	1.0	Weak foot contact	-1.0
Self contact (feet)	-0.4	Goal success burst	0.0
Ball contact hit	0.0	Approach stance	0.0
<i>Post-strike recovery posture</i>			
Stand height	-40	Stance asymmetry	-20
Stand orientation	-20	Feet width	-20
Knee width	-20	Yaw drift	-2
<i>Safety and regularisation</i>			
EE pos. divergence	-100	Feet air time	50.0
Swing feet clearance	-20	Excess contact force	-1.0
Hard landing	-1.0	Excess base lin. vel.	-1.0
Unstable	-0.5	Feet contact time	-0.5
No fly	-0.1	Lin. vel. z	-0.1
Undesired contacts	-0.1	Action smoothness	-0.1
Swing orientation	-0.04	Swing torso orient.	-0.04
Action rate L_2	-0.04	Arm default pose	-0.04
Feet slip	-0.02	Torque	-2e-7
Joint velocity	-5e-8	Strike joint acc.	-5e-8
<i>Alive</i>			
Kick alive	4.0		

Weights are as configured for the shooting stage. Each group is additionally multiplied by a group scale: motion tracking 2.0, shooting 0.5, recovery posture 1.0, safety 1.0, alive 1.0. The shooting scale is ramped linearly from 0 over the first 5000 iterations, and is held at 0 throughout the preceding motion-tracking stage, so the three zero-weight shooting terms and the whole shooting group are inactive until that stage boundary. The alive reward is attenuated by a factor 0.1 before ball contact, so it cannot be farmed by standing still and declining to strike. Terms listed here apply only in kick mode; locomotion-mode rewards are gated to exactly zero by the task-mode variable and are specified separately.

F. Domain randomisation

Randomisation broadens once at the Stage-1 to Stage-2 boundary, then holds constant. Beyond mass, centre-of-mass, friction, PD-gain and action-delay randomisation, two axes target a simulation-to-simulation gap on the strike: joint armature with internal friction, and the contact solver’s own response parameters. Standard humanoid randomisation perturbs the robot or a friction coefficient but leaves how contact is *resolved* fixed for all of training, which a policy can overfit for free.

V. EXPERIMENTAL SETUP

A. Platform and simulation

All experiments use a 29-DoF Unitree G1, policy at 50Hz emitting joint position targets, trained in IsaacLab [13] with thousands of parallel environments. Every reported simulation number comes from the exported deployment policy running in MuJoCo [18], not from training telemetry, so the cross-engine transfer gap sits inside the measurement rather than behind it.

B. Skill library

Seven kicking skills, each retargeted from a separate human video clip, are summarised in Table IV. Nominal bearings span 259.5° , from -121.74° to $+137.80^\circ$, at distances from 4.00 to 7.22 m; six strike with the right foot, one with the left. Bearing and distance follow from each skill’s configured ball and target placement, and the reachable band is the nominal bearing $\pm \theta_{\max}$, with $\theta_{\max} = 15^\circ$.

Coverage is reported as islands rather than a single span: the seven bands tile 155.8° across five disjoint islands, leaving gaps of 34.8° , 25.6° , 13.9° and 59.4° . Closing the largest needs $\theta_{\max} \approx 44.7^\circ$, a separate question from the one asked here.

C. Metrics

Shooting is scored by success at 0.5 m and 1.0 m, the nearest post-contact ball-to-target distance falling inside that radius, over all attempts so a whiff counts as failure. We also report shot error, peak ball speed and contact rate. Angular tolerance, the half-angle subtended by the 0.5 m radius at a given distance, is reported alongside, since the same radius is a harder target at 7.2 m than at 4.0 m.

Stability uses a strict fall criterion, not the tracking-based alive flag, which can fire while the robot is still standing. Locomotion generality is command-tracking error across the full envelope, terrain traversal and push recovery, all measured with the library attached.

D. Protocol

Unified-policy numbers are medians over a checkpoint sweep rather than single readings, because single-checkpoint evaluation of this task is unreliable: in our own data the checkpoint originally selected for $N=7$ scored 3.6% on skill_5 against a 36.6% median over 16 checkpoints from the same run, and per-skill ranges span up to 42 points. Each unified policy is swept over 8 checkpoints at 2000-step spacing (16 for $N=7$), 250 to 300 attempts per checkpoint. Teacher checkpoints are *not* swept: each is the specific artifact its student was distilled from, so its identity is fixed by the pipeline rather than sampled.

We summarise a sweep by its median rather than its mean because the per-checkpoint distributions are zero-heavy and right-skewed, so a mean is set by a few outlying draws. The choice does not carry any conclusion in this paper: under means the $N=7$ library average moves from 15.4% to 16.1% and the monotonic decline with library size is unchanged, while the reported fall rate roughly triples (0.6% to 1.9%). We report the estimator that is conservative on stability.

[Still to finalise: seed replicates per configuration, and the resulting replicate noise floor below which no effect is claimed. All results are currently single training runs.]

VI. RESULTS

Tables V, VI and VII report measured MuJoCo evaluations of the exported deployment policy. Unified-policy entries are medians over a checkpoint sweep (8 checkpoints, 16 for $N=7$); teacher entries are the single checkpoint each student

TABLE IV: Skill library geometry.

Skill	Foot	Dist. (m)	Nominal bearing	Reachable band ($\theta_{\max} = \pm 15^\circ$)	Ang. tol. @0.5 m
skill.1	right	5.17	-1.32°	$[-16.3, +13.7]$	$\pm 5.53^\circ$
skill.2	right	7.22	$+137.80^\circ$	$[+122.8, +152.8]$	$\pm 3.96^\circ$
skill.3	left	5.32	-121.74°	$[-136.7, -106.7]$	$\pm 5.37^\circ$
skill.4	right	5.99	-56.94°	$[-71.9, -41.9]$	$\pm 4.77^\circ$
skill.5	right	6.63	$+48.39^\circ$	$[+33.4, +63.4]$	$\pm 4.31^\circ$
skill.6	right	4.00	0.00°	$[-15.0, +15.0]$	$\pm 7.13^\circ$
skill.7	right	6.60	$+43.89^\circ$	$[+28.9, +58.9]$	$\pm 4.33^\circ$
Nominal span 259.5° ; reachable union 155.8° over five islands					

Bearing and distance are computed from each skill's configured ball and target placement; 0° is straight ahead and positive is to the robot's left. The reachable band is the nominal bearing plus or minus the trained aim range $\theta_{\max} = 15^\circ$. Angular tolerance is the half-angle subtended by the 0.5 m success radius at that skill's distance, so nearer targets are geometrically more forgiving. **Coverage is not continuous:** the seven bands leave four gaps of 34.8° , 25.6° , 13.9° and 59.4° ; closing the largest would require $\theta_{\max} \approx 44.7^\circ$. Six skills strike with the right foot and one (skill.3) with the left.

was distilled from. **Numbers still set in red (Table X) remain dummy scaffolding.**

A. Shooting quality

Table V compares against prior systems on the metrics they report. At matched forward geometry the skill.1 specialist reaches 44.3% success at 0.5 m with 0.69 m mean shot error, above every published figure in the table, so the locomotion-grounded substrate produces a competitive strike. The distilled seven-skill policy does not retain it: the same skill falls to 10.2%, and the library mean of per-skill medians is 15.4%. The full-library row has no prior counterpart, since no prior system strikes outside a forward cone. Prior success at 1.0 m is marked not reported rather than imputed.

B. Per-skill coverage

Table VI reports each skill separately; Fig. 3 shows the library geometrically. Panel (a) makes the five islands and the gaps between them visible, and adds something the table cannot: pace is direction-dependent, with forward strikes carrying two to three times the ball speed of the lateral and rearward ones. Panel (b) asks a different question, whether the strike survives the ball sitting where no skill was trained to expect it.

Stability and aim behave differently. Topple rates stay low across all seven bearings (mean 0.6%, worst case 2.6% on skill.7), so the locomotion substrate holds the robot up largely regardless of strike direction. Aim does not follow: success spans 0.8% to 36.6%. The weakest skills also fail differently. skill.2 contacts the ball 92.0% of the time and almost never places it, at the library's longest range (7.22 m, angular tolerance $\pm 3.96^\circ$, the tightest in the set), whereas skill.1 and skill.6 contact the ball on every attempt and still land at 10.2% and 7.6%, failing to aim rather than failing to strike. These call for different fixes.

C. Interference between skills

Table VII compares each skill's success in the unified policy against the teacher it was distilled from, across

four cumulative library sizes. **Interference is the system's dominant failure mode, and it scales with library size.** Mean success declines monotonically from 26.0% at $N=2$ to 15.4% at $N=7$, and the gap to the teachers widens from -5.9 to -17.7 points between $N=2$ and $N=6$ (-17.8 at $N=7$, over a different skill set, so the last two are not directly comparable).

That aggregate trend is the part four runs support. The per-skill picture is not orderly: skill.6 and skill.1 lose 34.2 and 34.1 points by $N=7$ while skill.4 loses 4.2, and individual skills move non-monotonically with N . Since each library is a separate distillation run, per-skill scatter still cannot be separated from run-to-run variance without seed replicates. What the scaling series does establish is that the loss is not an artifact of one bad merge: it is present at every library size and grows with it.

The skill that collapses is the one the geometry predicts. skill.2 asks for the library's most extreme rotation ($+137.8^\circ$) and was distilled from one of its weakest teachers (19.6%, second lowest of the seven), and it ends up the weakest skill in the library: a 0.8% median, at or below 1.2% at 10 of the 16 swept checkpoints and exactly 0.0% at 4 of them. Two checkpoints nonetheless reach 22.0% and 30.0%, so the behaviour is representable by this architecture and reachable within this run. The collapse is therefore a capacity result rather than a broken skill definition or a mis-specified aim convention: the hardest geometry paired with a thin teacher is where the merge runs out of capacity first.

What the comparison does establish is that the loss occurs in the merge and not in the substrate: every specialist was trained from the same Stage-1 base, and several reach success rates the distilled policy does not approach.

D. Handoff

Table VIII evaluates the final unified policy on both directions of the mode boundary. Kick-to-locomotion is scored on survival alone, since the return deliberately prevents contact, and survives both the policy's own auto-return at the end of the clip and an abort forced at an arbitrary tick. Locomotion-to-kick is scored on survival, contact and aim, and we sweep the entry speed rather than report a single in-motion condition, because contact saturates near 100% on most skills and a contact-only reading cannot detect an entry that strikes just as reliably while aiming worse.

Entering the kick while moving costs both stability and accuracy, and the cost grows smoothly with entry speed. Scaling the walk-command envelope from zero to the full locomotion training range moves contact 95.3 to 84.1, success at 0.5 m 17.0 to 11.9, success at 1.0 m 32.7 to 25.7, and topple 0.7 to 4.2, monotonically on all four. Measured against its own zero-command control, full-speed entry scores lower on 15 of the 16 checkpoints (sign test $p = 0.0005$). The transition itself is not the fragile part, since the robot survives 95.8% of full-speed entries, but momentum is not free: the accuracy it costs is of the same order as the accuracy the library has to give. What the measurement does not cover is the approach, since the ball

is placed relative to the robot rather than waited for, and that is where we locate the remaining precondition in Sec. VII.

E. Locomotion generality

Table IX is the axis prior systems do not report. The comparison is paired against the same substrate trained without any skill layer, and the claim is an absence of effect: attaching seven skills should not degrade the locomotion underneath. That claim needs the replicate noise floor before a small difference means anything; each measured Δ is small next to the MAE it is taken against. Terrain traversal is scored on the same sustained-fall criterion training itself uses (10 consecutive ticks below the height threshold, not an instantaneous dip): an earlier version of this measurement used an instantaneous check and reported rough terrain failing 94% of the time, which traced entirely to ordinary gait oscillation crossing the threshold for a tick or two before recovering, not to any loss of balance.

F. Ablations

Table XI isolates two design choices on a single specialist, skill.7’s Stage-B checkpoint, rather than the deployed library. Both mechanisms are configured identically at Stage B and Stage C (the post-strike safety reward is not gated by the shooting-reward switch, and the pose-injection probabilities are byte-identical between the two stages’ configs), so Stage B isolates the mechanism itself from library interference (Table VII) and from the aim-conditioned shooting objective, neither of which either mechanism depends on. This is a different checkpoint from the Stage-C teacher used elsewhere in the paper (Tables V–VII): Stage B’s ball observations are held at zero by construction, so the policy strikes without aim feedback, appropriate for isolating balance recovery but not comparable to the specialist’s shooting numbers.

The first sub-table compares post-strike stabilisation mechanisms, scored on settled-stance quality rather than survival: a forced-abort survival measurement scores the arms almost entirely in the phase and task mode the ablated reward is switched off in, which an earlier pass of this table initially (and misleadingly) reported. Scored correctly, the shaped arm holds the lowest settled error on nearly every checkpoint and is the only one that settles at all over the recovery window, though the result is narrower than it first appears and we bound it explicitly with the table. The second isolates the source of kick-disturbed recovery poses, evaluated on two distributions so that a policy trained on reference-clip poses is also tested on real post-kick states. That one is decisive too: the gap between an arm’s two columns widens monotonically as its training distribution moves away from the live one, exposing exactly the gap between motion capture and what the policy produces at contact.

G. Hardware

Table X reports real-robot trials, with the simulation-to-hardware gap on the same metric rather than the hardware number alone. (a) covers shooting quality, including a commanded-approach condition that exercises the

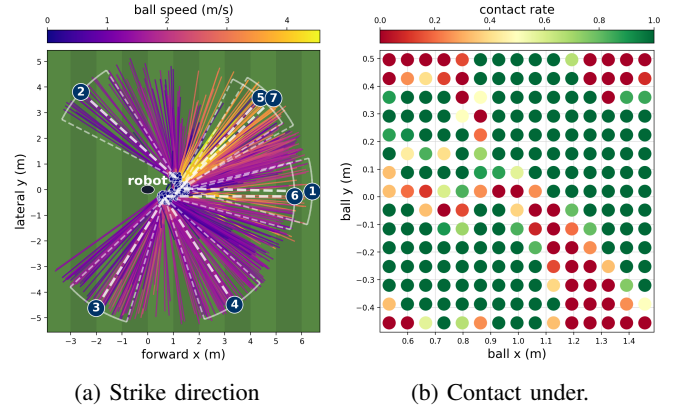


Fig. 3: **What the library covers, and how far the ball may sit from where a skill expects it.** (a) Every ball contact from 100 trials on each of the seven skills (564 of 700; the remainder never touched the ball), with the ball jittered ± 20 cm about each skill’s own trained spawn, twice the ± 10 cm used in training. Dashed rays mark each skill’s nominal bearing and the dashed outlines its commanded $\pm 15^\circ$ aim band; colour is ball speed. The seven skills resolve into five islands, since skills 1/6 and 5/7 overlap, and forward strikes carry two to three times the pace of the lateral and rearward ones, a difference no aggregate speed can express. Trajectories spill well outside their aim bands: that is the aim bias quantified in Sec. VII, not commanded spread. Contact is not accuracy, and these are contacts. (b) Contact rate of the *nearest* trained skill across 1500 ball placements spanning the union of all seven trained spawns (15×15 bins, 70.4% overall). Most of that region lies outside training: 82% of placements sit further from any skill’s spawn than the ± 10 cm jitter it trained under, median 20 cm, maximum 60 cm. Contact survives displacement far better than aim does, since only about a third of contacts anywhere in this region become directed shots, and the nearest-skill assignment grants an oracle choice that a deployed selector would have to make for itself. The two panels are separate experiments: (a) jitters each skill about its own spawn, (b) sweeps one shared region across all of them.

locomotion-to-kick handoff under operator-issued velocity commands rather than a scripted approach. (b) is the one sim result from Table VIII we consider it dishonest to leave unvalidated on hardware: kick-to-locomotion survival under a mid-swing abort. We report it phase-stratified rather than as a random-tick statistic, since a human-operated trigger cannot reproduce the uniform-random sampling the sim measurement uses.

VII. LIMITATIONS AND CONCLUSION

A. Limitations

Distillation loses most of the teachers’ accuracy, and the loss grows with library size. This is the system’s principal open problem. Mean success at 0.5 m falls from 33.2% across the teachers to 15.4% in the merged seven-skill policy, and skill.1, the forward skill that still beats every

TABLE V: Shooting quality in context. Prior-work numbers are as reported by each paper; evaluation geometry differs, so read the bearing and distance columns alongside the results.

Method	Bearing range	Dist. (m), tol.	suc.@0.5 m (%)	suc.@1.0 m (%)	Shot err. (m)	$v_{\text{ball}}^{\text{max}}$ (m/s)	Hit (%)	Alive (%)
PPO [29]	forward	n/r	0.0±0.0	0.0±0.0	4.721±.007	0.780±.003	0.0±0.0	0.0±0.0
AMP [11]	forward	n/r	0.9±0.1	3.3±0.1	3.733±.004	1.771±.051	38.6±2.8	41.6±2.3
PAiD [3]	forward	n/r	8.2±0.7	19.5±0.5	1.850±.026	4.986±.017	67.7±0.3	67.3±0.5
RoboNaldo [2], stage 2, free-kick	forward, ±38.7°	5.0, n/r	28.8±1.3	65.5±0.6	0.899±.012	14.792±.011	82.1±0.3	100.0±0.0
RoboNaldo [2], stage 3, free-kick	forward, ±38.7°	5.0, n/r	36.3±0.1	66.5±0.2	1.055±.012	14.178±.034	81.1±0.1	99.7±0.1
RoboNaldo [2], stage 3, moving-ball	forward, ±38.7°	5.0, n/r	32.4±0.6	63.3±0.6	1.131±.004	13.875±.004	79.2±0.1	98.8±0.1
Ours , loco-grounded specialist, skill_1	≈forward, ±15°	5.17, ±5.5°	44.3±16.8	77.5±12.1	0.69±.17	3.67±.15	100.0±0.0	100.0±0.0
<i>Multi-directional library: a broader capability envelope (Table I)</i>								
Ours , unified $N=7$, library	259.5° span, 155.8° union	4.0–7.2 ±4.0° ±7.1°	15.4±12.7	32.2±21.9	1.68±.76	2.86±.77	96.7±5.9	99.4±1.0

Only the upper block is like-for-like: every row there addresses a single forward cone, and our specialist – the same locomotion-grounded substrate this paper builds on, at $N=1$, not a separately tuned forward-kicker – is the most accurate in it, though over a narrower aim range than RoboNaldo’s (±15° vs ±38.7°), which the bearing and tolerance columns expose. The lower block is a different capability envelope rather than a worse entry in the same context: it alone accepts arbitrary velocity commands, strikes off-axis and rearward, and hands back to a trained locomotion controller both ways (Table I), paying for that breadth in per-skill accuracy. Bold marks the best value in a column among all rows shown, error columns taken low-is-best. n/r = not reported (angular tolerance is never stated in this literature). Hit is ball-foot contact, independent of whether the ball then reaches the target; shot error and peak speed are computed only over hits, so a low-Hit row’s sample for those two columns is smaller and skewed toward its easier attempts. RoboNaldo defines its own Contact column the same way (the ratio of environments that make ball contact), so the two are the same quantity, not merely similarly named. Bearing range: other rows address one contiguous arc, so span equals coverage; our seven skills span 259.5° but tile only 155.8° of it, in five disjoint islands (Table IV). Baseline and RoboNaldo figures, std included, are as reported in RoboNaldo’s [2] own Table 2; earlier drafts of this table marked suc.@1.0m not reported because we had not yet located that source directly. RoboNaldo’s simulated targets lie on a goal plane 5 m ahead, giving ±38.7° of azimuth entirely within the forward hemisphere; its hardware numbers are taken at 3 m under a different protocol and are not mixed into this table. We list both of their reported stages: stage 2 is trained on free-kick only, stage 3 adds locomotion and moving-ball shooting on top of it, and is evaluated on both regimes. Their free-kick success improves from stage 2 to stage 3 (28.8% → 36.3% at 0.5 m) even though stage 3 is not specialised to that condition, the opposite direction from the interference we measure when merging skills (Table VII). RoboNaldo’s own paper does not state whether its ± is computed across training seeds or evaluation episodes, so we cannot claim protocol parity with it. Ours are precisely defined but differ between the two rows. The specialist row is one teacher checkpoint, and its ± is the standard deviation across 100 evaluation batches of 10 attempts. The library row is a median over 16 checkpoints of seven skills, and its ± is the spread across those seven per-skill medians (0.8% to 36.6%, Table VI), so it measures how unevenly the library performs rather than evaluation noise on any one skill. Neither is variance across independently trained seeds, which Sec. V still leaves open.

We report the specialist and the distilled policy separately because they differ substantially. At matched forward geometry the specialist reaches 44.3% at 0.5 m, above every published figure in this table, so the locomotion-grounded substrate produces a competitive strike. The deployed library policy does not retain it, at 15.4% averaged over all seven skills; Table VII gives the per-skill breakdown, including skill_1’s own drop to 10.2%. Sec. VII treats this as the system’s principal open problem rather than presenting the specialist as the deployed result.

published figure we compare against (44.3%), is among the worst affected, dropping to 10.2%. The scaling series makes clear this is not one unlucky merge: the deficit is already present at $N=2$ (−5.9 points) and widens monotonically as skills are added. The substrate and the per-skill training are therefore not the bottleneck: the merge is. What we cannot yet say is whether the mechanism is a capacity limit of a single network or something correctable in the distillation procedure itself, since individual skills move non-monotonically across library sizes and we have not run seed replicates. Until that is resolved, the multi-skill claim this paper makes is about coverage and stability, not about retaining specialist accuracy.

The strikes are precise but mis-aimed, which is where the shot error comes from. Regressing the ball’s post-contact departure bearing on the commanded aim offset θ (a slope of 1 would mean the strike steers with the command) shows the aim channel is only weakly realised. Across checkpoints the slope sits near 0.2–0.9 rather than 1 for most skills, in the specialists as much as in the merged policy, and each skill carries a systematic angular offset of roughly 10–30° while per-trial scatter about its own bias is only 5–8°. The kicks are therefore repeatable; they are aimed consistently in the wrong direction. This matters for reading Tables V and VI: the shot-error and success columns are dominated by that per-skill bias rather than by imprecision,

which is why hit rate and survival stay high while success at 0.5 m does not. It also gives the distillation gap above a concrete instance rather than only an aggregate: skill_1’s specialist is the one teacher that does steer reliably (slope 1.07, range 0.85–1.52 over four checkpoints), and the merged policy does not inherit it (slope 0.26, range −0.21–0.80 over eight), acquiring a +29° bias the teacher did not have. We do not claim this generalises: of the two other teacher/student pairs we measured this way, one is unchanged and one steers *better* after merging. Whether a per-skill angular calibration would recover most of the lost accuracy is untested, and is the first thing we would try.

Coverage is not continuous. The seven bands tile 155.8° across five islands, leaving four gaps of up to 59.4°; a bearing inside a gap has no skill that addresses it. Closing the largest needs $\theta_{\text{max}} \approx 44.7^\circ$, which raises a question we do not settle: whether wider commanded aim conflicts with the reference clip, since the target rotates and the clip does not. Adding clips near the gap centres is the alternative, trading library size against aim range.

The handoff survives, but entry speed costs accuracy, and the approach into it is unmeasured. Kick-to-locomotion is robust to an abort at an arbitrary tick. Locomotion-to-kick survives at every entry speed we test, 95.8% alive even at the full locomotion command range, but accuracy falls monotonically with the velocity carried into

TABLE VI: Per-skill shooting performance and coverage.

Skill	Suc.@0.5 (%)	Suc.@1.0 (%)	Shot err. (m)	v_{ball}^{max} (m/s)	Hit (%)	Alive (%)
skill_1	10.2	22.0	1.90	3.63	100.0	99.6
skill_2	0.8	3.2	2.92	2.22	92.0	100.0
skill_3	20.6	40.4	0.98	2.84	85.2	98.6
skill_4	25.8	51.0	1.10	2.82	100.0	100.0
skill_5	36.6	67.4	0.81	2.04	100.0	100.0
skill_6	7.6	23.8	1.89	2.36	100.0	100.0
skill_7	6.4	17.8	2.15	4.13	100.0	97.4
mean	15.4	32.2	1.68	2.86	96.7	99.4

All rows are the final unified policy carrying all seven skills, evaluated one skill at a time; they are not per-skill specialists. The specialist comparison is Table VII. Foot, nominal bearing and target distance are per-skill geometry, not measurements, and are reported once in Table IV rather than repeated here. **Every entry is a median over 16 checkpoints** spanning 387k to 444k training steps, 250 attempts per checkpoint (4000 per skill), rather than a single checkpoint; Sec. V explains why. Success is nearest post-contact ball-to-target distance within the stated radius, counted over all attempts so that a whiff is a failure. Alive uses the same strict fall criterion as Tables V and VIII, not the tracking-based flag that can fire while the robot is still standing. No prior system reports per-skill directional coverage, so this table has no baseline rows.

Aim is highly non-uniform across the library, spanning 0.8% to 36.6%, and the weak skills fail in different ways. skill_2 is the clearest failure: it contacts the ball 92.0% of the time but almost never places it, at the library’s longest range (7.22 m) and tightest angular tolerance ($\pm 3.96^\circ$). skill_1 and skill_6, both near-forward, stay in single digits to low double digits (10.2% and 7.6%) despite contacting the ball on every attempt, so their loss is aiming rather than striking. Contact is now at or near 100% for five of the seven skills, which isolates aim as the binding constraint across the library. Alive rates are high across the board but not uniform: skill_7 (97.4%) and skill_3 (98.6%) carry nearly all the instability, while four of the seven skills never fall.

the strike: success at 0.5 m drops from 17.0% at a zero-command entry to 11.9% at full range, and contact from 95.3% to 84.1%. Survival is therefore not the fragile part of this direction; aim under momentum is, and a deployment that triggers kicks while the robot is still moving should expect to pay for it. A second limitation sits one step earlier. Every locomotion-to-kick trial in Table VIII places the ball for the robot, at the correct robot-relative offset at the trigger instant, because a ball left where it lay would be kicked by the random walk itself and would confound a handoff failure with a trip. Nothing in that table therefore asks a moving robot to arrive at a ball that stays put. Approaching under the commandable controller and then triggering is the composition the architecture rests on, and the hardware commanded-approach condition in Table X is the only place we probe it at all. What does not hold is any claim that a kick can be triggered at an arbitrary velocity and connect with a ball the robot had to walk to first.

Skill selection is commanded, not learned. Which skill fires is an input, not a decision the policy makes. Our critique of handcrafted triggering in prior work applies to our own system here.

Perception is out of scope. The policy consumes a ball position in the heading frame and does not estimate it. Training models perception degradation through noise, latency, per-episode bias and dropped readings, but we report no perception experiments and make no claim about the estimator supplying that input on hardware.

TABLE VII: Negative transfer: specialist vs. unified.

Skill	Teacher	$N=2$	$N=4$	$N=6$	$N=7$
skill_1	44.3	37.8 (−6.5)	22.5 (−21.8)	29.0 (−15.3)	10.2 (−34.1)
skill_2	19.6	14.2 (−5.4)	8.0 (−11.6)	5.3 (−14.3)	0.8 (−18.8)
skill_3	31.8	—	31.8 (+0.0)	19.2 (−12.6)	20.6 (−11.2)
skill_4	30.0	—	20.7 (−9.3)	8.8 (−21.2)	25.8 (−4.2)
skill_5	46.4	—	—	30.2 (−16.2)	36.6 (−9.8)
skill_6	41.8	—	—	15.3 (−26.5)	7.6 (−34.2)
skill_7	18.4	—	—	—	6.4 (−12.0)
mean	33.2	26.0 (−5.9)	20.8 (−10.7)	18.0 (−17.7)	15.4 (−17.8)

Success at 0.5 m in percent, with the change from that skill’s own teacher in parentheses. Libraries are cumulative (skill_1 through skill_7), so a dash means the skill is not in that library. Unified entries are medians over 8 swept checkpoints (16 for $N=7$); each mean-row delta is taken over the teachers of that library’s own skills, so the four deltas are not differences of a common baseline.

The teacher column is a single checkpoint by construction, not a median: that specific checkpoint is the artifact distillation consumed, so sweeping it would measure policies that were never teachers.

skill_2’s near-zero is a measured value, not a missing one: it is exactly 0.0% at 4 of the 16 swept checkpoints and at or below 1.2% at 10, with the two largest draws at 22.0% and 30.0%. Reporting the mean instead of the median would raise it to 4.8% on the strength of those two draws, with a spread of ± 8.8 around it.

Interference grows with library size. Mean success falls monotonically ($26.0 \rightarrow 20.8 \rightarrow 18.0 \rightarrow 15.4$) and the gap to the teachers widens ($-5.9 \rightarrow -10.7 \rightarrow -17.7$, then -17.8 at $N=7$ over a different skill set). The per-skill picture is far less orderly: skill_6 and skill_1 lose 34.2 and 34.1 points by $N=7$ while skill_4 loses only 4.2, and several skills move non-monotonically with N (skill_4: $-9.3, -21.2, -4.2$). Since each library is a separate distillation run, that per-skill scatter still cannot be separated from run-to-run variance without seed replicates; the aggregate trend is the part these four runs support.

B. Conclusion

We presented a humanoid soccer policy built the other way around from current practice: a commandable locomotion policy trained first and alone, with N motion-guided kicking skills layered on top as task-gated additions whose objectives are held at exactly zero outside kick mode. Because that gate is a hard zero rather than a reduced weight, the reachable gait space is fixed by the command curriculum rather than by any reference clip; because every skill begins and ends in the same locomotion state, a library of N skills needs $O(N)$ transitions rather than $O(N^2)$.

The instantiation carries seven skills spanning 259.5° , including lateral, rearward and weak-foot strikes no single forward-facing reference expresses, and replaces the scripted post-strike stabilisation tail with a handoff to the trained locomotion controller. Alongside shooting quality we report command-tracking, terrain and push-recovery measured with the full library attached: the axis prior systems do not report, and the one that tests whether locomotion survives being used as a substrate.

[Final quantitative summary sentence, once measurements replace the scaffolding in Tables V to XI.]

TABLE VIII: Handoff robustness, both directions.

Direction	Trigger condition	Alive	Hit	Suc.@0.5	Suc.@1.0
Kick→Loco	end of authored clip	99.1	n/a	n/a	n/a
Kick→Loco	random tick, incl. strike [†]	98.8	n/a	n/a	n/a
Loco→Kick	from $v \approx 0$ state	99.3	93.3	15.4	30.3
Loco→Kick	in motion, 25% range	97.2	90.2	14.7	29.8
Loco→Kick	in motion, 50% range	97.0	87.8	13.7	27.5
Loco→Kick	in motion, full range [‡]	95.8	84.1	11.9	25.7

All values are percentages. Reported on the final unified policy carrying all seven skills, not a controlled ablation arm. Medians over the same 16 checkpoints as Tables V–VII, each rate pooled over all seven skills (1750 trials per checkpoint for the clip-end and settled rows, 2240 for the random-tick row, 672 for each in-motion row, including the zero-command control described below though it is not shown as its own row). The settled row anchors the ball to the robot’s own frame at the trigger; an earlier version fixed it in the world frame before a 1.5 s settle over which the robot drifts about 4 cm backward, which left the ball systematically off-nominal and understated that row. Kick→Loco is scored on survival alone, since the return deliberately prevents further ball contact: the first row is the policy’s own auto-return at the end of the authored clip, the second an abort forced at an arbitrary tick, including inside the strike window. Loco→Kick is scored on both survival and whether the entry actually produces a strike.

[†]An earlier version of the random-tick row drew that tick uniformly from one fixed window (ticks 10–60 since trigger, mirroring training’s own `kick_abort_delay` default) that, verified against each skill’s own scrubbed strike boundaries, never actually reaches the strike phase for any of the seven skills: the earliest strike start among them is tick 82. That version’s own 100.0% therefore measured only the approach, the easiest phase to abort from, not the row it was labelled as. The corrected measurement instead draws the tick from three ONNX-derived sub-windows per skill (pre-strike, strike, post-strike), giving the strike window equal representation with the combined non-strike windows rather than none; its strike-only component, 98.1%, is the same number Table XI(b)’s live-donor column reports on the deployed library, and the two are now directly comparable because they share the same tick-selection mechanism.

[‡]The three rows below the settled one are one experiment: the walk command envelope scaled to 25, 50 and 100% of the locomotion training range ($v_x, v_y \in [-1, 1]$ m/s, $\omega_z \in [-1, 1]$ rad/s), one command drawn per trial and held fixed for a 2 to 3 s window before the trigger. All three share one scan with the same ± 10 cm ball jitter, the same $\pm 15^\circ$ aim sweep and the same robot-frame ball anchoring as each other. That scan also has a fourth, 0%-range arm not shown as a table row (alive 99.3, contact 95.3, success 17.0 and 32.7): its purpose is as a control for the three shown, not as a fifth trigger condition, since it differs from the settled row above by scan rather than by speed (a real gap on its own, 17.0 against 15.4 at zero speed). Comparing the full-range row directly against the settled row would confound entry speed with that scan-to-scan gap; comparing it against this control isolates speed instead, and paired checkpoint by checkpoint the full-range row scores lower on 15 of 16 against it (sign test $p = 0.0005$). An independent 5-iteration repeat of the full-range condition reproduces the table’s own values closely (83.0 contact, 11.9 and 24.4 success, 95.8 alive). One defect in an earlier version of this scan is worth recording, because it inverted the conclusion. The scan set the policy’s velocity command before each step, but the deployed policy re-reads that command from its own controller input within the same step and overwrites it; with no controller attached the command was silently zeroed and the robot stood still. Every in-motion number we measured before the fix was in fact the zero-command row above, which is precisely why entering in motion appeared to cost nothing.

One difference remains, and it bounds what these rows support. The ball is teleported to its robot-relative offset at the trigger instant, because a ball left in place would be struck during the walk itself and would confound a fall of the handoff with a trip over the ball. The entry is therefore never asked to *arrive* at a ball that stays where it is, only to strike one already in front of it, and the walk command is random rather than ball-directed, so the robot is usually strafing, reversing or turning rather than closing on the ball. These rows establish that the transition survives and strikes while the robot is moving, not that a moving robot positions itself correctly; Sec. VII treats the latter as the open question it is.

TABLE IX: Locomotion generality with the skill library attached.

Command axis / task	Loco-only	With library ($N=7$)	Δ
v_x forward MAE (m/s)	0.39	0.42	+0.02
v_x backward MAE (m/s)	0.51	0.55	+0.04
v_y lateral MAE (m/s)	0.35	0.35	+0.00
ω_z yaw MAE (rad/s)	0.21	0.25	+0.04
Terrain traversal (%)	98.7	100.0	+1.3
Push recovery (%)	100.0	100.0	+0.0

Paired comparison against the same locomotion substrate trained without any skill layer. The claim is *no degradation*, so the relevant question is whether each Δ falls inside the seed-replicate noise floor. Every Δ here is a few hundredths, small next to the MAE itself. Each MAE is the mean per-tick ℓ_1 error summed over both commanded linear axes (v_x and v_y together for the two v_x rows, v_x and v_y together for the v_y row, against a fixed 0.8 m/s / 0.5 m/s / 0.5 rad/s step command), not a residual purely on the named axis, so a walking gait’s ordinary per-step lateral sway on a nominal-zero cross axis contributes to a same-axis-labeled row; this does not affect the paired comparison, since both columns share the same definition. Terrain traversal uses the same sustained, not instantaneous, fall check training itself uses (10 consecutive ticks below threshold, see source comment); an earlier instantaneous version manufactured an apparent 94% failure rate on rough terrain out of ordinary, self-recovering gait oscillation. No prior humanoid soccer system reports this axis.

TABLE X: Hardware results.

(a) Shooting trials.					
Condition	Trials	Suc. [†] (%)	Err. (m)	Contact (%)	Alive (%)
Static ball, forward	30	50.0	0.25	93.3	100.0
Static ball, lateral	30	46.7	0.19	90.0	100.0
Static ball, rearward	30	16.7	0.87	93.3	96.7
Commanded approach	30	36.7	0.30	93.3	100.0

(b) Handoff robustness, Kick→Loco, manual abort.			
Abort phase	Trials	Alive (%)	
Pre-strike	20	95.0	
Mid-strike (contact)	20	95.0	
Post-strike/recovery	20	100.0	

The simulation-to-hardware gap is reported on the same metric rather than only the hardware number in isolation.

(a) Static conditions place the ball at a taped, measured offset from the robot with no approach, and group skills in the library by how far their nominal bearing lies from the forward axis: forward, skill_1 and skill_6 ($\approx 0^\circ$); lateral, skill_4 and skill_5 ($\approx 45\text{--}49^\circ$ either side); rearward, skill_2 and skill_3 ($\approx 120\text{--}140^\circ$). Each row is 30 trials. A row summarises its direction class, not any one skill: members differ substantially in simulation (Table VI). Commanded approach starts the robot away from the ball and drives it in with operator-issued velocity commands (walk/turn keys or joystick, not the autonomous ball-approach mode) across every skill except skill_7 (skill_1 through skill_6), so this row specifically exercises the locomotion-to-kick handoff from a general commandable state rather than a scripted approach, matching Table VIII’s Loco→Kick, settled condition.

[†]Success radius is distance-corrected, not a flat 0.5 m. The room this hardware was tested in only allows the target tape to sit 1.42–2.08 m from the ball, seven times closer than the 4.0–7.2 m simulated skills train and are scored at (Table IV). Holding the 0.5 m radius fixed while shrinking the distance would silently widen the effective angular tolerance to $13\text{--}20^\circ$, three to five times looser than the $\pm 4\text{--}7^\circ$ every other table in this paper scores against, so a flat 0.5 m here would not be measuring the same thing it measures elsewhere. Each row instead uses $r = d_{\min} \tan(\bar{\theta})$, where $\bar{\theta}$ is the mean of Table IV’s own angular tolerance over that row’s pooled skills and $d_{\min} = 1.42$ m is the room’s nearest tape distance: forward $\bar{\theta}=6.33^\circ \Rightarrow r=0.158$ m; lateral $\bar{\theta}=4.54^\circ \Rightarrow r=0.113$ m; rearward $\bar{\theta}=4.67^\circ \Rightarrow r=0.116$ m; commanded approach $\bar{\theta}=5.18^\circ \Rightarrow r=0.129$ m. Per-trial target distance was not logged, so the true radius for any given trial lies somewhere in $[d_{\min} \tan \bar{\theta}, d_{\max} \tan \bar{\theta}]$ for $d_{\max} = 2.08$ m, the room’s farthest tape distance; anchoring at $d_{\min}$ is the conservative end of that bracket; it never inflates the reported success rate, since every trial shot farther than 1.42 m is graded against a tighter-than-necessary radius rather than a looser one. Err. is the mean nearest ball-target distance over contacted trials only, matching this paper’s shot-error convention elsewhere (a whiff has no shot to grade).

(b) Aborted via the deployment stack’s manual return-to-locomotion control (keyboard 1 / joystick RB+D-pad down), which can be pressed at any point during the swing. The operator judged and triggered each phase by eye against the robot’s own motion, timed to land before ball contact (pre-strike), at the moment of contact (mid-strike), or during the post-contact recovery window (post-strike). This is *not* the same protocol as Table VIII’s sim result, which samples the abort tick uniformly at random via script. We report phase-stratified survival rather than reusing the word “random” for a claim a human trigger cannot support. Mid-strike is the condition Table VIII identifies as riskiest in sim (single-support, momentum-carrying); the robot was spotted for these trials.

TABLE XI: Ablations, measured on skill_7’s Stage-B checkpoint (single specialist, not the deployed library).

(a) Post-strike stabilisation mechanism.			
Arm	Stance asym. ↓ rad ²	Yaw drift ↓ (rad/s) ²	Orient. tilt ↓ sin θ
Posture + contact safety (ours)	0.013	0.012	0.003
Contact safety only	0.276	0.074	0.163
No stabilisation	0.117	0.044	0.073

(b) Kick-state pose injection for locomotion recovery.		
Training arm	Alive, ref.-clip (%)	Alive, live donor (%)
Kick-state transplant (ours)	88.4	81.8
Reference-clip pose only	82.3	59.7
No pose injection	71.8	32.1

Every arm is now measured. (b)’s entries are the median over 8 checkpoints (250k–300k, evenly spaced), 25 iterations $\times$ 32 trials per checkpoint; (a)’s are the median over the same 8 checkpoints, 10 iterations $\times$ 16 trials. All five arms resume from the same model_0250000, so they share a common ancestor and an identical evaluation protocol; (b)’s shared 250k checkpoint scores identically (50.9%) on its own measurement across all three of its arms, which is what a correct protocol should produce. (b)’s live-donor column exercises the same kick-to-locomotion flip mechanism Table VIII reports on, since a pose the policy’s own dynamics produced mid-strike is exactly what a live donor is, but **the two tables’ alive numbers are not comparable and should not be read against each other**. Three things differ at once. This column reports the strike-window subset alone, the hardest phase to abort from, where Table VIII’s random-tick row pools every abort tick; it measures a Stage-B specialist whose ball observations are held at zero, not the deployed library; and it draws from 250k–300k rather than 387k–444k. To isolate which of those differences drives the gap, we ran the same split-by-strike-phase measurement on the deployed unified checkpoints Table VIII itself sources, holding the eval type, the strike-window definition, and split-vs-pooled fixed, and changing only what this ablation already varies: specialist versus library, and checkpoint range. There, strike-window alive rate is 98.1% (median over the same 16 checkpoints; worst single checkpoint 91.6%), close to Table VIII’s own pooled random-tick figure of 98.8% and far from this ablation’s 81.8%; that same 98.1% is in fact the strike-only component underlying Table VIII’s row, the two having been measured by the same sweep. The remaining gap is therefore attributable to what the checkpoint already differs on – an early, single-skill Stage-B network against a fully-trained seven-skill deployment – and not to a broken protocol or a disagreement between the two tables’ own measurements.

(a) is measured differently from (b), and differently from an earlier version of this table – see below for why. Survival alone does not appear as a column: under (a)’s own measurement (an uninterrupted strike, clip played to completion, no forced abort), all three arms reach at or above 99% survival, indistinguishable from one another. The three columns shown are settled-stance error magnitudes, unweighted, in the units the underlying reward terms themselves use (*stance asymmetry*: summed squared left/right joint-angle deviation; *yaw drift*: squared base yaw angular velocity; *orientation tilt*: $\sin \theta$ for θ the base’s tilt from vertical, obtained as the horizontal magnitude of the unit gravity direction expressed in the base frame, hence dimensionless, and deadzoned at 0.025), averaged over the settled window described below. The three are different physical quantities and are not comparable across columns; only the column-wise comparison between arms is meaningful.

Both sub-tables are measured on skill_7’s Stage-B checkpoint, not the deployed library: both mechanisms are trained identically at Stage B and Stage C, so this isolates the mechanism from library interference and from the aim-conditioned shooting objective, which neither mechanism depends on. This checkpoint is not the Stage-C teacher used in Tables V through VII; its ball observations are held at zero by construction, so it strikes without aim feedback and its numbers here are not comparable to that specialist’s shooting performance.

(a) ablates two reward families: the six-term post-strike recovery posture group, and the in-strike contact-safety terms (excess foot-contact force, hard landing, excess base velocity). “Contact safety only” keeps the latter and drops the former, so it is penalised for dangerous contact but given no shaping toward a stable recovered stance. The gait-shaping terms that share the safety reward group (foot air time, swing clearance) are held fixed in every arm, so no arm loses its gait for reasons unrelated to stabilisation.

An earlier version of (a) scored the wrong phase, and the corrected measurement reverses the conclusion. We first scored this ablation on survival under a forced mid-strike abort (Table VIII’s own kick-to-locomotion flip mechanism) and found the posture group scored *worse* than dropping it. Two properties of the mechanism being ablated make that measurement invalid for this question. The posture-shaping reward is gated to zero throughout locomotion-approach and the strike itself, fading in only after the clip passes into its post-swing content; a mid-strike abort scores

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